

Cadence: The Theory of Everything

How our smart anklet spots a Parkinson's "freeze" and buzzes you free — explained from scratch.

ENGF0031 Scenario 2 • Smart Clothing • Every number in here is from our own real ankle recording capture1.csv.

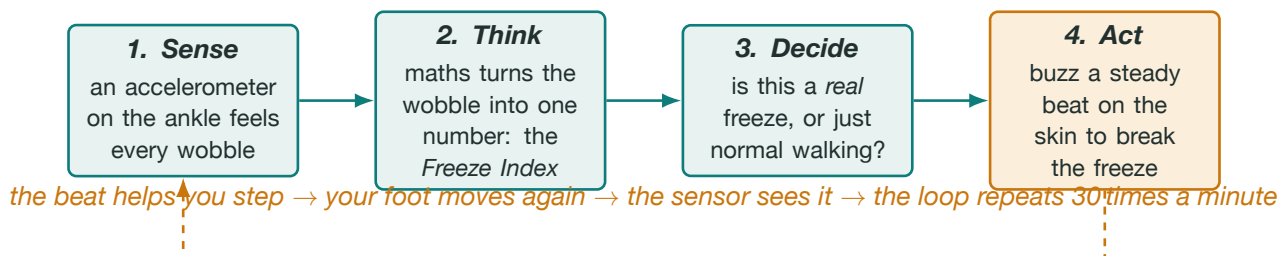
This guide explains *everything* the project does and *why*, in plain language, with pictures. The last section is the honest "things that broke and how we fixed them" diary. No prior physics needed.

1 The big idea: a thermostat for your feet

People with Parkinson's disease sometimes get **freezing of gait (FoG)**: their feet feel "glued" to the floor even though they want to walk. It is a major cause of falls. A weird but well-proven trick helps: give the person a **steady external beat** — a metronome tick, a line on the floor, or a gentle buzz — and their brain "borrows" that rhythm and starts stepping again. This is called *rhythmic cueing*.

Our garment, **Cadence**, is a small computer worn on the ankle that does this automatically. It watches how your leg is moving, decides when you have frozen, and buzzes a beat *only then*. When you start walking again, it goes quiet. That "watch → decide → act → watch again" circle is called a **closed loop**.

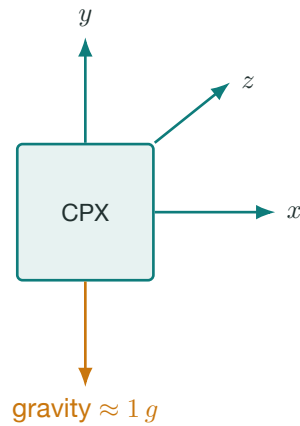
In plain English: It is a thermostat, but for walking. A thermostat senses the temperature and flips the heater on only when the room gets cold. Cadence senses your *leg*, and flips a *buzzer* on only when your walking "freezes".



2 Step 1 — Feeling movement: the accelerometer

The sensor is an **accelerometer**: imagine a tiny weight on tiny springs inside the chip. When the chip speeds up, slows down, or tilts, the weight lags and the springs stretch, and the chip measures that stretch. It reports three numbers — one for each direction in space: *x* (side to side), *y* (up and down), *z* (forward and back). Gravity always tugs the weight "down", so even when you are still the sensor reads about **1 g** (one gravity) pointing at the floor.

We sample these three numbers **64 times every second** (64 Hz). That is fast enough to see a leg tremble but slow enough for a coin-cell-powered chip.



We don't care *which* way your ankle is rotated, so we squash x, y, z into a single “how hard is it shaking” magnitude:

$$a = \sqrt{x^2 + y^2 + z^2}$$

In plain English: Strap the chip to your ankle however you like — we only use the *total* shake $a = \sqrt{x^2 + y^2 + z^2}$, so it works no matter how the strap is twisted. That trick is called being *orientation-invariant*.

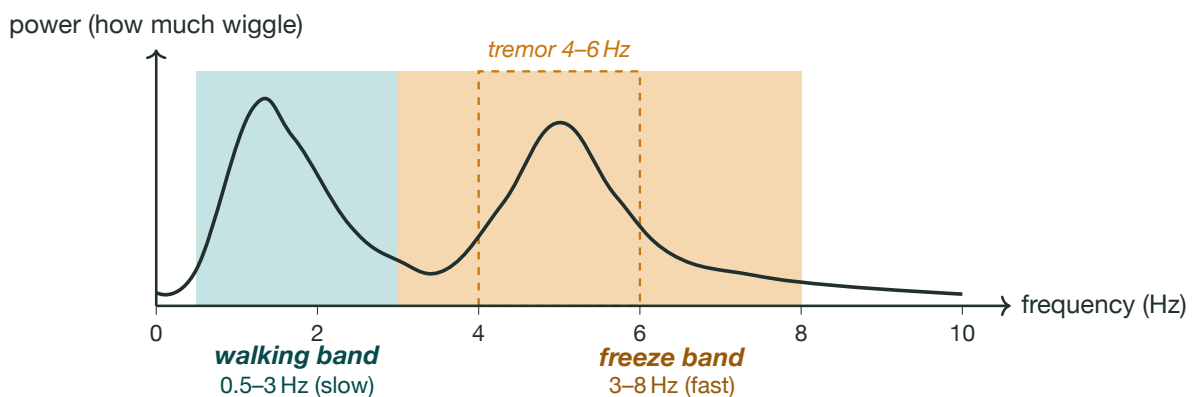
3 Step 2 — The magic number: frequency and the Freeze Index

Here is the key insight the whole project rests on. ***Walking and freezing shake your leg at different speeds of wiggle.***

- **Walking** is a slow, big swing — about 1 to 2 steps per second. We call that a low *frequency* (0.5–3 Hz), the “walking band”.
- **Freezing** is a fast, small *tremor* — the leg vibrates roughly 3 to 8 times per second while going nowhere. That is a high frequency (3–8 Hz), the “freeze band”.

In plain English: Frequency just means *how many wiggles per second*. Think bass notes (slow, deep) versus treble notes (fast, high). Walking is the bass; a freeze tremor is the treble.

To measure “how much slow wiggle vs. how much fast wiggle” is in a chunk of signal, we use a standard maths recipe (a *power spectrum*). You can picture it as the bar-graph equaliser on a music app: it tells you how loud each frequency is in the mix.



(Illustrative shape of a spectrum. Walking puts a big bump in the slow band; a freeze adds a bump in the fast band.)

Now the clever bit. **Moore et al. (2008)** defined the **Freeze Index (FI)** as a simple ratio:

$$\text{Freeze Index} = \frac{\text{power in the FAST freeze band (3–8 Hz)}}{\text{power in the SLOW walking band (0.5–3 Hz)}}$$

- **Walking:** lots of slow power, little fast power $\Rightarrow$ *small* FI (a small number on top divided by a big number on the bottom).

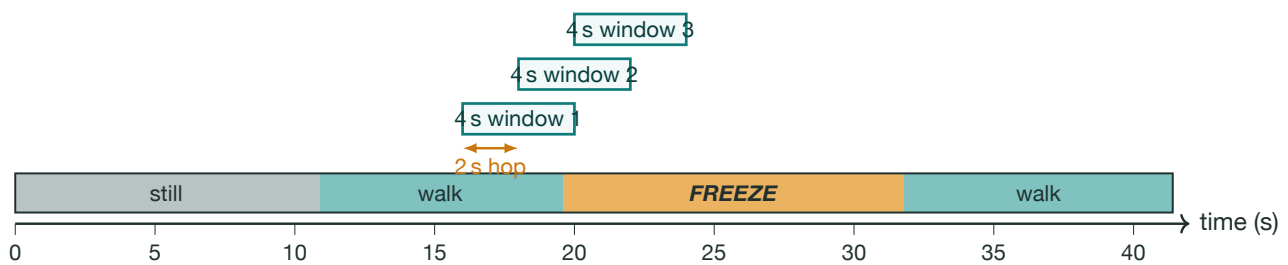
- **Freezing:** the leg trembles fast but doesn't travel, so fast power shoots up and slow power collapses $\Rightarrow$ *big FI*.

One number tells the whole story. A low Freeze Index means “moving normally”. A high Freeze Index means “shaking in place” — the signature of a freeze. Everything after this is just deciding *how high is high enough*.

4 Step 3 — Chopping time into windows

We can't compute a frequency from a single instant — you need to watch the leg for a little while to see how fast it's wiggling. So we slice the stream into **4-second windows**, and we slide the window along by **2 seconds** each time, so windows overlap by half. That gives us a fresh decision **every 2 seconds** while still having 4 seconds of context to measure frequency accurately.

In plain English: Like reading a sentence with a 4-word-wide highlighter that you shuffle forward 2 words at a time — you always see enough words to understand, and you re-read often enough to keep up.



(The coloured strip is our real recording. The overlapping boxes are the sliding analysis windows.)

5 Step 4 — The decision, with two safety nets

A naive detector would just say “Freeze Index above some line $\Rightarrow$ buzz”. But that buzzes at the wrong times. Two famous failure modes, and how we block each one:

The threshold: where is “the line”?

From our real recording, normal **walking** peaked at a Freeze Index of **2.08**. A real **freeze** hit **6.6, 21.4, 5.3...** far higher. So we set the line at **FI > 2.10** — just above the worst walking value, so ordinary walking can never trip it. (The textbook starting point is 2.0; we *tuned* ours to 2.10 to fit our own data.)

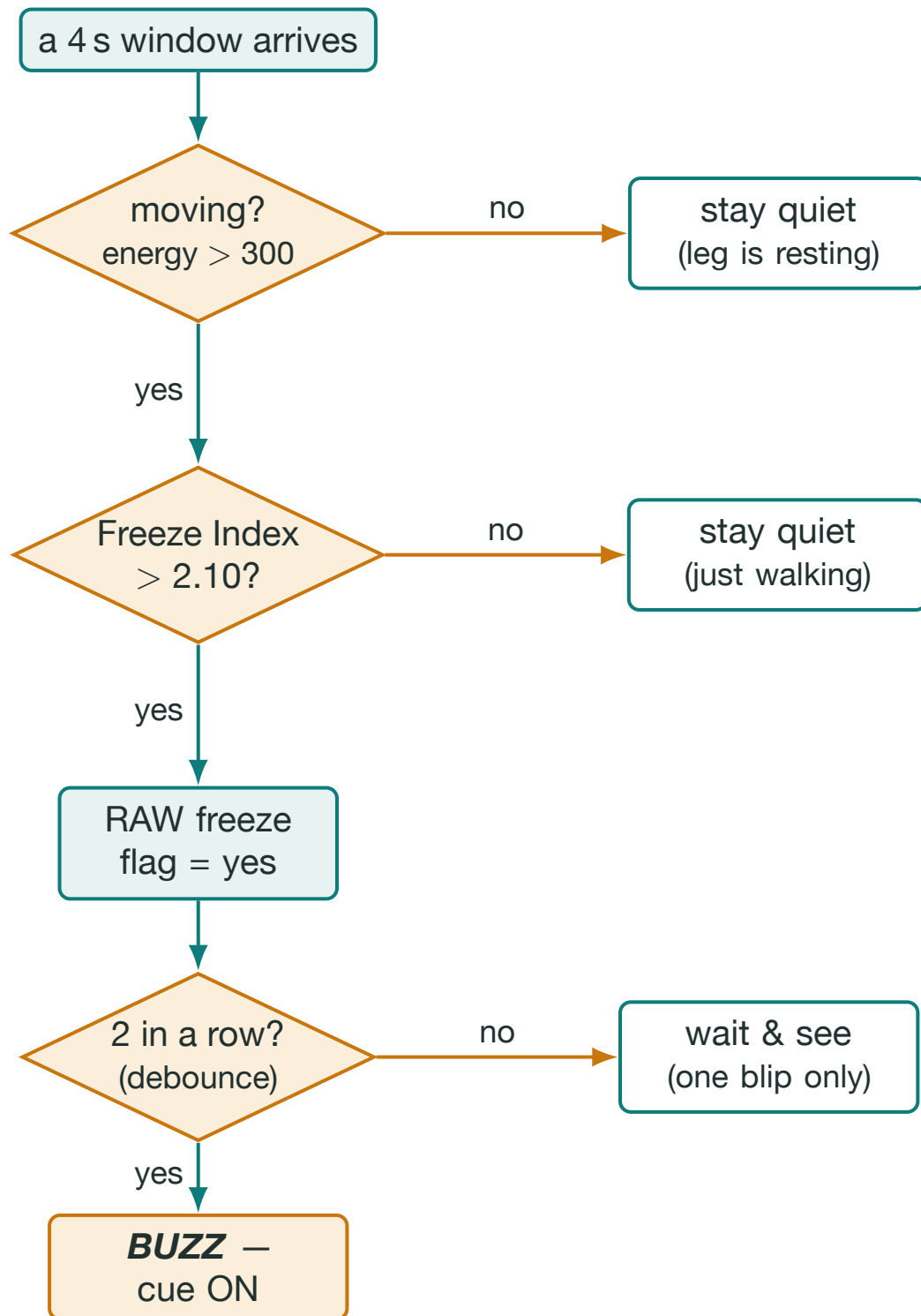
Safety net #1 — the energy gate (don't shout at a resting leg)

A ratio can be misleading. If the leg is *completely still*, both bands are just tiny sensor noise — and noise over noise can randomly look “high”. So before we even look at the Freeze Index we ask: **is the leg actually doing anything?** We add up the real movement power; if it is below **ENERGY_FLOOR = 300** (a resting ankle reads only ~ 130 – 270), we declare “not moving” and stay silent no matter what the ratio says.

Safety net #2 — debounce & hysteresis (don't trust one blip)

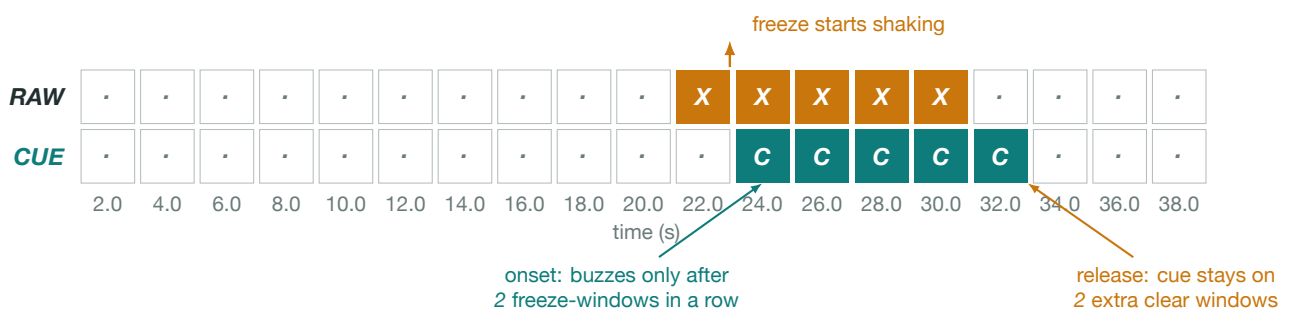
A single noisy window shouldn't fire the motor. So we *debounce*: the buzzer only switches **ON after 2 freeze-windows in a row**, and only switches **OFF after 2 clear windows in a row**. Requiring a run before changing state is called *hysteresis*.

In plain English: Like waiting to feel a second raindrop before opening your umbrella — and leaving it open until you're sure the rain has really stopped.



Watching the safety nets work, second by second

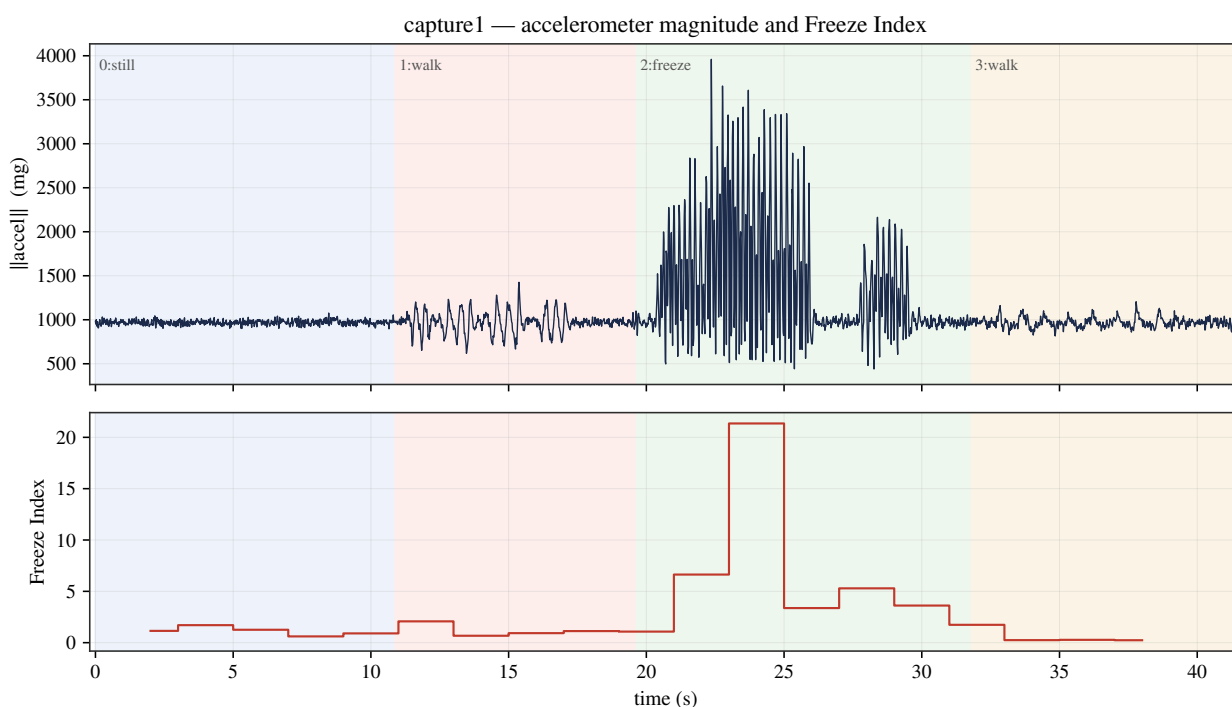
Here is the actual run over our recording. Row 1 is the raw “this window is over the line” flag; Row 2 is what the motor actually does after debouncing.



Notice three things: (1) the cue starts *one window later* than the raw flag — that is the 2-in-a-row onset rule; (2) it *stays on* for one window after the shaking stops — the 2-clear release rule, so it doesn't flicker; (3) the very first freeze window (t=20s) didn't even raise the raw flag, because that window straddles the walk→freeze boundary and is half-walking. We catch the freeze 2 s later anyway.

6 Step 5 — Did it actually work? The real scoreboard

We recorded one continuous ~41-second trip on a real ankle: **still** → **walk** → **freeze** → **walk**. Here is the real trace the detector saw (top: how hard the ankle shook; bottom: the Freeze Index, with our 2.10 line):



To grade it we compare, window by window, what the detector said against the truth. Four outcomes are possible, shown in a **confusion matrix**:

		detector says...	
		Buzz	Quiet
truth	real freeze	TP = 5 <i>caught it</i>	FN = 1 <i>missed</i>
	really not	FP = 0 <i>false buzz</i>	TN = 13 <i>correctly quiet</i>

The two headline numbers (this is what the worksheet grades — not “accuracy”):

- **Sensitivity** = $\frac{TP}{TP + FN} = \frac{5}{6} = 83\%$ — “of all the real freezes, how many did we *catch*?” (Don’t leave the patient stuck.)
- **Specificity** = $\frac{TN}{TN + FP} = \frac{13}{13} = 100\%$ — “of all the non-freezes, how many did we correctly *leave alone*?” (Don’t annoy them with wrong buzzes.)

Why not just say “accuracy”?

Because freezes are *rare*. In a long walk, a lazy detector that buzzes *never* would still be “correct” most of the time — high accuracy, but totally useless (it catches zero freezes). Splitting the score into **sensitivity** (catch rate) and **specificity** (leave-alone rate) exposes that trick and is the honest way to report a medical detector. Our one miss (FN) is the very first 2-second slice of the freeze, which the next window catches — so the wearer is still cued, just ~ 2 s later.

7 The honest part: everything that broke, and how we fixed it

Real engineering is mostly fixing things that don’t work the first time. Here is our actual debugging diary — the bugs, *why* each happened, and the fix.

A. Getting the data off the board

Bug 1 — a flaky USB cable kept dropping mid-recording

What happened. Half-way through a capture the serial connection died and the script crashed, losing the run.

Why. The only cable we had was failing intermittently — the laptop briefly lost the board, and our code assumed the connection never disappears.

The fix. We made the reader *reconnect automatically*: if the port vanishes it waits, re-opens it, and keeps reading. The next capture survived a drop and still produced one clean 41-second file.

Bug 2 — we were “flying blind” (no live readout)

What happened. While recording, nothing printed, so we couldn’t tell if data was flowing.

Why. Python *buffers* its output — it holds text back until a big chunk piles up, instead of showing each line.

The fix. We ran Python in *unbuffered* mode (the `-u` flag), so every sample echoed instantly and we could watch the capture live.

B. Tuning the brain (the parameters we “fixed” to real values)

Bug 3 — normal walking nearly triggered a false freeze

What happened. With the textbook line at $FI = 2.0$, a stride of *normal walking* hit $FI = 2.08$ and almost set off the buzzer.

Why. Our walking happened to be a touch shakier than the textbook assumed, so its Freeze Index crept above 2.0.

The fix. We *tuned the threshold up to 2.10*, just above the worst real walking value. Result: walking now *never* fires (specificity 100%), and real freezes (FI 3–21) still sail over the line.

Bug 4 — a perfectly still leg could randomly look “frozen”

What happened. When the leg was motionless, the Freeze Index sometimes spiked for no real reason.

Why. The Index is a *ratio* of two tiny noise signals; dividing noise by noise gives garbage that can look high.

The fix. We added the *energy gate* (safety net #1): if total movement is below 300, declare “not moving” and stay silent regardless of the ratio. A freeze is *trembling while trying to move* — a still leg can’t be freezing.

Bug 5 — single noisy windows caused jittery buzzing

What happened. One odd window could flick the motor on (or off) for a moment — annoying and unconvincing.

Why. We were reacting to every individual window with no memory.

The fix. We added *debounce* + *hysteresis* (safety net #2): need 2 freeze-windows in a row to start, and 2 clear windows in a row to stop. The cue is now smooth and only responds to *sustained* freezes.

Bug 6 — the detector “missed” the first freeze window (a false negative)

What happened. The window centred at $t = 20$ s (right as the freeze began) scored a low FI and was marked “no freeze”.

Why. A 4-second window that lands on the walk→freeze boundary is *half walking, half freezing*, so its average Freeze Index is watered down.

The fix & trade-off. Real freezes last several seconds, so the very next window ($t = 22$ s) catches it and the cue fires — only ~ 2 s late, which is clinically fine. We *accept* this one miss to keep false buzzes at zero. (A neural-net upgrade, FoGNet, is the “given more time” way to shave it further.)

C. Building the documents (LaTeX & shell gremlins)

Bug 7 — the worksheet PDF wouldn’t compile

What happened. Two LaTeX errors: a maths line with tangled $\backslash (\dots \backslash)$ and $\$ \dots \$$ markers, and an “undefined command” for $\| \cdot \|$.

Why. The maths delimiters were mismatched, and the symbol needed the `amsmath` package which wasn’t loaded.

The fix. We rewrote the line cleanly and added `\usepackage{amsmath}`. It compiled.

Bug 8 — a code block split across two pages, and a shell glob blew up

What happened. A 20-line code listing broke awkwardly across a page break; separately, a `rm ws-*.png` command aborted the whole pipeline.

Why. LaTeX placed the page break mid-listing; and the `zsh` shell errors out on `*` when *no* files match the pattern.

The fix. We forced a `\newpage` so the listing stays whole, and used `find ...-delete` instead of a bare `*` glob so an empty match is harmless.

8 One-page cheat sheet

Accelerometer	tiny weight-on-springs chip; reports shake in x, y, z at 64 times/second. We use the total magnitude $\sqrt{x^2 + y^2 + z^2}$.
Frequency	how many wiggles per second. Walking = slow (0.5–3 Hz); freeze tremor = fast (3–8 Hz).
Freeze Index	= fast-band power $\div$ slow-band power. Low = walking, high = freezing. Our trigger line: > 2.10 .
Energy gate	ignore the leg unless it's truly moving (energy > 300). Stops a still leg from faking a freeze.
Debounce	need 2 freeze-windows in a row to buzz, 2 clear to stop. Kills jitter.
Window	4 s of data, re-checked every 2 s (50% overlap).
Sensitivity	catch rate of real freezes = 83% (5 of 6).
Specificity	leave-alone rate of non-freezes = 100% (13 of 13).
Closed loop	sense $\rightarrow$ think $\rightarrow$ decide $\rightarrow$ buzz $\rightarrow$ you move $\rightarrow$ repeat. A thermostat for walking.

Built for ENGF0031 Scenario 2 “Smart Clothing”. All performance numbers come from our own real ankle recording (capture1.csv, 2652 samples over 41.4s). The spectrum sketch is illustrative; every other figure reflects real captured data.